

6. (3 points)

Given the following information for an insurer's books of business as of December 31, 2017:

Accident Year	Onlevel Premium (\$000)	Cumulative Paid Loss (\$000)	Estimated Reserves (\$000)
2014	1,000	275	400.00
2015	1,200	306	553.85
2016	1,500	344	818.18
2017	1,700	220	1,133.33

- The estimated reserves for all accident years are calculated using the Cape Cod method.
- The expected loss payment pattern is approximated by the following loglogistic function where G is the cumulative proportion of ultimate losses paid and x represents the average age of paid losses in months: $G(x) = \frac{x}{x + \theta}$.

a. (1.5 points)

Calculate the expected loss ratio used in the Cape Cod method.

b. (0.75 point)

Evaluate the appropriateness of using the Cape Cod method for this book of business.

c. (0.75 point)

Briefly describe the two types of variance associated with a statistical model for loss reserving. Identify an approach to reduce one of the types of variance.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 6	
TOTAL POINT VALUE: 3	LEARNING OBJECTIVE(S): A2, A3
SAMPLE ANSWERS	
Part a: 1.5 points	
<u>Sample 1</u> Reserve = Prem × ELR × (1 – lag) = Prem × ELR × [1 – G(x)] $1 - G(x) = \Theta / (x + \Theta)$ $400 = 1,000 \times \text{ELR} \times [\Theta / (42 + \Theta)]$ $553.85 = 1,200 \times \text{ELR} \times [\Theta / (30 + \Theta)]$ $0.4 \times (42 + \Theta) = \text{ELR} \times \Theta$ $0.4615 \times (30 + \Theta) = \text{ELR} \times \Theta$ $0.4 \Theta + 16.8 = 13.845 + 0.4615 \Theta$ $2.955 = 0.0615 \Theta$ $\Theta = 48.05$ $\text{ELR} = 0.4 \times (42 + 48.05) / 48.05 = 0.75$ <u>Sample 2</u> Reserves = Adj. Prem × ELR × (1 – G(x)) $1,133.33 / 818.18 = [1,700 \times \text{ELR} \times (1 - G(6))] / [1,500 \times \text{ELR} \times (1 - G(18))]$ $1.385 = 1.133 \times [(1 - G(6)) / (1 - G(18))]$ $1.222 = [\Theta / (6 + \Theta)] \times [(18 + \Theta) / \Theta]$ $7.332 + 1.222 \Theta = 18 + \Theta$ $0.222 \Theta = 10.668$ $\Theta = 48.05$ $G(42) = 42 / (42 + 48.05) = 0.466$ $G(30) = 0.384$ $G(18) = 0.273$ $G(6) = 0.111$ $\text{ELR} = (275 + 306 + 344 + 220) / (1,000 \times 0.466 + 1,200 \times 0.384 + 1,500 \times 0.273 + 1,700 \times 0.111)$ $\text{ELR} = 0.751$ <u>Sample 3</u> $1,000 \times \text{ELR} \times [1 - G(42)] = 400$ $1,200 \times \text{ELR} \times [1 - G(30)] = 553.85$ $1,500 \times \text{ELR} \times [1 - G(18)] = 818.18$ $1,700 \times \text{ELR} \times [1 - G(6)] = 1,133.33$ $0.4 = \text{ELR} \times [1 - 42 / (42 + \Theta)]$ $0.4615 = \text{ELR} \times [1 - 30 / (30 + \Theta)]$	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

$$1.15375 = [1 - 30 / (30 + \Theta)] / [1 - 42 / (42 + \Theta)]$$

$$1.15375 = [\Theta / (30 + \Theta)] \times [(42 + \Theta) / \Theta]$$

$$1.15375 = (42 + \Theta) / (30 + \Theta)$$

$$1.15375 \times (30 + \Theta) = 42 + \Theta$$

$$34.6125 + 1.15375 \Theta = 42 + \Theta$$

$$0.15375 \Theta = 7.3875$$

$$\Theta = 48$$

$$0.4 = \text{ELR} \times [1 - 42 / (42 + 48)]$$

$$0.4 = \text{ELR} \times 0.5333$$

$$\text{ELR} = 0.75$$

Sample 4

$$\text{ELR} = \text{Paid Loss} / \text{Used-up Prem} = 1,145 / p \quad (p \text{ equals used-up premium})$$

$$\text{On Level Prem} = 1,000 + 1,200 + 1,500 + 1,700 = 5,400$$

$$\text{Total Reserves} = 400 + 553.85 + 818.18 + 1,133.33 = 2,905.36 = \text{ELR} \times (5,400 - p)$$

$$= 2,905.36 / (5,400 - p)$$

$$1,145 / p = 2,905.36 / (5,400 - p)$$

$$1,145 \times 5,400 - 1,145p = 2,905.36p$$

$$1,145 \times 5,400 = 4,050.36p$$

$$1,526.5 = p$$

$$\text{ELR} = 1,145 / 1,526.5 = 75\%$$

Part b: 0.75 point

Sample 1

AY	Avg Age	(1) G(x)	(2) Cum Pd Loss	(3) = (2)/(1) Ultimate Loss	(4) Premium	(3)/(4) Ultimate Loss Ratio
'14	42	0.467	275	588.865	1,000	0.589
'15	30	0.385	306	794.805	1,200	0.662
'16	18	0.273	344	1,260.073	1,500	0.840
'17	6	0.111	220	1,981.928	1,700	1.166

↓ upward trend

The loss ratio is increasing from AY to the next. It is not appropriate to use Cape Cod.

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample 2

Testing for ELR constancy

AY	Avg Age	G(x)	Used-up Premium	LR
2014	42	0.4668	466.77	0.5892
2015	30	0.3847	461.657	0.6628
2016	18	0.2728	409.215	0.8406
2017	6	0.1112	188.959	1.1643

$\Theta = 47.98$ (from part a)

$G(x) = x / (x + \Theta)$

$G(42) = 42 / (42 + 47.98) = 0.4668$

Used up prem = On Level Prem $\times$ G(x)

LR = Cumulative Paid loss (given) / used up prem

The ELR does not seem appropriate because the loss ratios (LR) are increasing indicating that reserves will be overstated in older years and understated in more recent years.

Part c: 0.75 point

Sample 1

Process variance – the variance associated with the random chance associated with modelling the insurance process

Parameter variance – the variance associated with the chance that the parameters of our model are incorrect

Parameter variance can be reduced by limiting the number of parameters in our model.

Sample 2

Process variance – risk from the randomness of insurance business

Parameter risk – risk that estimated parameters do not correctly predict insurance losses, and don't fully account for uncertainty

To reduce parameter risk, use Cape Cod over LDF method because Cape Cod uses additional information, thus less parameters

Sample 3

Process variance – due to the pure randomness of the valuation

Parameter variance – due to that we can't exactly estimate the parameters

Reduce parameter variance: try to reduce parameter's number. The fewer parameters, we will have smaller parameter variance

EXAMINER'S REPORT

Candidates were expected to know how to calculate the expected loss ratio used in the Cape Cod method, and be able to evaluate the appropriateness of using the Cape Cod method. They were also expected to know the two types of variance and know the basics behind both.

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part a

Candidates were expected to know the formula for estimated reserve. Given this knowledge, the question provided information to set up four equations for the estimated reserve by accident year. Using two of these equations there would be two unknowns (Theta and ELR). The candidate then had two ways to solve for these: 1. Divide one equation by the other to eliminate one of the variables (presumably ELR), or 2. Substitute one equation into the other (presumably solve for ELR in one equation and substitute it into the other equation). Once you solve for Theta, you can use one of these two equations to solve for ELR.

Common mistakes included:

- Not knowing how to solve for the two unknowns (Theta and ELR) once the equations had been set up, or using a method of solving which was accurate but perhaps more time consuming (e.g. quadratic equation).
- Calculating $G(x)$ as Paid Loss / (Paid Loss + Estimated Reserve) even though the problem stated $G(x) = x / (x + \Theta)$ where x represents the average age of paid losses.

Part b

Candidates were expected to know how to translate $G(x)$ into either expected ultimate loss using the LDF method or used up premium, and then see if there was a trend in loss ratios to evaluate the appropriateness of using a constant Expected Loss Ratio across all accident years for this book of business.

The most common error was adding the Cape Cod reserve to paid loss to calculate the ultimate loss for each accident year. Candidates were expected to recognize that they were asked to evaluate the appropriateness of using a constant ELR for all accident years combined and that the given reserves were already estimated with the Cape Cod method, using the constant ELR from part a.

Other mistakes included:

- Stating that the Cape Cod method is not appropriate because the premium volume is increasing.
- Not using the growth function provided in the question.

Part c

Candidates were expected to know basic information about both process variance and parameter variance and be able to give a definition of each.

Common mistakes included:

- Only listing the two types of variance without including a definition.
- Saying that using fewer parameters will reduce process variance.